Modeling electrical losses in CsPbI3 solar cells: stacked idealizations and a routine-quality envelope at 18.0%

Hussain Touhid Siddiquee* Md. Abdul Malek Fahim

Abstract

Stacked idealisations inflate SCAPS-1D CsPbI3 projections. Using SCAPS-1D on one fully worked ITO/TiO2/CsPbI3/CBTS/Ni stack, joint optimisation of absorber thickness, doping, bulk defects, gap, and both transport-layer affinities gives a permissive-box ceiling of **24.95%** (Voc 1.321 V, Jsc 21.20 mA/cm^2, FF 89.11%; 2.2 μm absorber, N_A 3×10^{16} cm^{-3}, N_t 10^{12} cm^{-3}, χ_{CBTS} 3.9 / χ_{TiO2} 3.7 eV, 10^{10} cm^{-2} interface defects on both heterojunctions). Sampling a frozen routine-quality prior volume (96-point Latin hypercube over eight factors with published, measured-anchored ranges) gives a prior-predictive envelope of median **18.0%** with 5–95% range **15.6–20.0%**, above which the 2024–2026 experimental records fall. Every load-bearing idealisation is priced individually, including radiative recombination, mobility, series resistance to 5 Ωcm^2, contacts, affinities, optics, and density of states, for which the ceiling-versus-record comparison is reported both ways. The defended number is the 20.82% routine-defect result, not the ceiling: the defect floor sits beyond the best reported films and both optimum affinities lie outside surveyed values.

Keywords: CsPbI3; perovskite solar cell; SCAPS-1D; CBTS; reproducibility; device simulation

I INTRODUCTION

All-inorganic CsPbI3 combines improved thermal stability with a bandgap near 1.7 eV, suitable for single-junction and tandem top-cell use [1, 2]. Buried-interface engineering carried laboratory efficiencies past 20% in 2024 [3] and to 22% in 2025–2026: 22.60% laboratory and 22.02% certified by moisture-responsive crystallisation [1], 22.05% by fluorinated interfacial dipoles [2], with supporting records of 21.71% with dual-interface and bulk co-passivation at 1.27 V [4] and 21.43% certified for inverted cells [5]. The 22.02% certified value of Dai et al. is the highest certified CsPbI3 value cited here. Each advance came from defect and interface control.

SCAPS-1D is widely used for pre-experimental screening. The ITO/TiO2/CsPbI3/CBTS stack of Hossain et al. [6] progressed from 17.90% over 96 transport-layer combinations [7] to 19.06% after optimisation. Later simulation reports reach 24.17% on the same stack [8] and 24.24% on ZnO/PTAA without interface layers [9]. Three recurring problems accompany this inflation. First, one-variable-at-a-time optimisation hides interactions among thickness, doping, defects, gap, and alignment. Second, interface and defect assumptions are often incompletely reported. Third, scripted SCAPS sweeps can mix unit systems and silently alter defect populations.

A survey of more than 250 SCAPS perovskite architectures attributes much of this inflation to unrealistic inputs, especially the zero radiative-recombination default, unphysical defect floors and mobilities, isolated optimisation, and optical defaults [10]. A companion optical-loss framework implements the wavelength-dependent half of that protocol [11]. This paper implements the electrical-loss half on one stack: factorial screening, priced radiative, mobility, resistance, contact, affinity, and density-of-states assumptions, and a routine-quality sensitivity envelope.

*Department of EEE, Leading University, Sylhet-3112, Bangladesh. E-mail: touhidsiddiqueeraj@gmail.com, mdabdulmalekfahim@gmail.com.

This work contributes a five-factor guardrailed screen and joint affinity map locating the ceiling at 24.95%; priced single-assumption corrections; stacked realistic variants; a measured-grade calibration ladder; a 95-run routine-quality envelope with a separate radiative sensitivity subset; interface dose–response in portable lifetime/velocity metrics; and a range-dependent Morris analysis.

II LITERATURE: CsPbI3 PROGRESS AND SIMULATION INFLATION

Black-phase CsPbI3 progressed from 10.8% quantum-dot films through γ-films and graded-gap devices [12–14]. Compositional engineering continued this rise [15, 16]. The 18.4% β-phase result then stood until buried-interface work reached 20.6%, 20.1% stabilised [3, 17]. The 2025–2026 interface wave is shown in Figure 1.

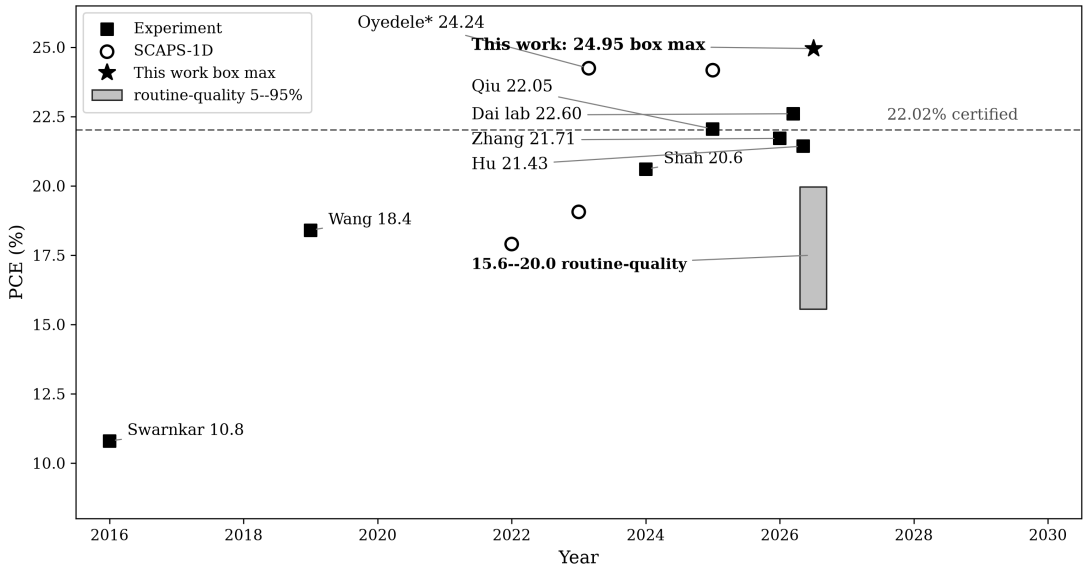


FIG. 1: CsPbI3 record race. Open symbols are simulations; the star and band are this work's ceiling and envelope.

The simulation studies trace a parallel inflation curve. The CBTS programme reached 17.90% and 19.06% [6, 7], the re-parameterised baseline used here. Later claims report higher values on different stacks or with unclear interface treatment. Because several published parameter tables are incomplete, other groups' cells are compared as reported, not re-placed on our curve.

The hole transport layer facilitates hole extraction while suppressing interface recombination. Cu2BaSnS4 (CBTS) is a wide-gap ($\sim$1.9 eV) inorganic hole conductor whose valence alignment with CsPbI3-class absorbers suits it to the p-side of the stack, and it is the hole transport material of the baseline programme reproduced here [6, 7]. The electron transport layer must extract electrons at the perovskite–ETL interface while blocking holes; TiO2 is the established choice for this stack, reported as the superior ETL with the CBTS hole transport material in CsPbI3 devices [6].

In this study, SCAPS-1D simulation is used to investigate one fixed stack, ITO/TiO2/CsPbI3/CBTS/Ni, under factorial variation of absorber thickness, doping, bulk and interface defect densities, gap, both transport-layer affinities, temperature, parasitics, and density of states. The output metrics (Voc, Jsc, FF, PCE) of each configuration are evaluated against interface-inclusive guardrails, and the surviving configurations are reported as a permissive-box ceiling, a routine-quality envelope, and a defended routine-defect result.

TABLE I: Selected CsPbI3 results: experiment versus simulation. Our entry is flagged, not ranked: ⋆permissive-box ceiling; the defended envelope median is 18.0% with 5–95% range 15.6–20.0%. Certified values are marked ^{c}; other experimental values are laboratory-reported.

Study	Stack	Type	PCE (%)	IF
Swarnkar 2016	TiO2/Spiro/MoOx	Exp.	10.8 [12]	–
Wang 2019	TiO2/Spiro	Exp.	18.4 [17]	–
Shah et al. 2024	TiO2/2D-Cs2PbI2Cl2	Exp.	20.6 (20.1 stab.) [3]	buried-passivated
Qiu 2025	CsPbI3 + DFAz dipole	Exp.	22.05 [2]	passivated
Zhang 2026	TiO2 dual-interface+bulk	Exp.	21.71 [4]	co-passivated
Dai 2026	TiO2/PTABr/Spiro	Exp.	22.60 lab / 22.02^c [1]	passivated
Hu 2026 (Joule)	inverted HPDA	Exp.	21.43^c [5]	2D-stitched
Lei 2025	AP-modified CsPbI3	Exp.	22.16 [18]	passivated
Xu 2026	QD-interface, inverted	Exp.	22.15 [19]	QD-passivated
Hossain 2022	TiO2/CBTS	Sim.	17.90 [7]	on
Hossain 2023	TiO2/CBTS/Ni	Sim.	19.06 [6]	on
Nazli 2025	TiO2/CBTS/Ni	Sim.	24.17 [8]	unclear
Oyedele 2023	ZnO/PTAA	Sim.	24.24 [9]	OFF
This work⋆	TiO2/CBTS/Ni	Sim.	24.95	on

III NUMERICAL SIMULATIONS

A SCAPS-1D model

SCAPS-1D solves Poisson's equation with electron and hole continuity under drift-diffusion transport [20–22]:

$$-\frac{\partial}{\partial x}\left(\varepsilon(x)\frac{\partial V}{\partial x}\right) = q[p - n + N_D^+ - N_A^- + p_t - n_t], \tag{1}$$

$$\frac{\partial n}{\partial t} = \frac{1}{q}\frac{\partial J_n}{\partial x} + G_n - R_n, \quad \frac{\partial p}{\partial t} = -\frac{1}{q}\frac{\partial J_p}{\partial x} + G_p - R_p, \tag{2}$$

$$J_n = q\mu_n nE + qD_n\nabla n, \quad J_p = q\mu_p pE - qD_p\nabla p, \tag{3}$$

with Shockley–Read–Hall recombination through the specified bulk and interface defect levels. Optical generation uses the Eg-sqrt model, $\alpha = A\sqrt{h\nu - E_g}$, so gap sweeps move the absorption edge rather than using a frozen spectrum.

B Device structure and re-parameterised baseline

Devices were simulated under AM1.5G illumination at 100 mW/cm^2 and 300 K unless stated, using SCAPS-1D 3.3.10 with scripted sweeps (four parallel workers) and Python (matplotlib) post-processing for analysis and figures. The starting point reproduces Hossain et al. [6]: ITO, 30 nm TiO2, CsPbI3, 100 nm CBTS, and Ni. Neutral interface recombination blocks are present at both heterojunctions. The absorber uses the edge-moving absorption model. This is therefore a re-parameterised baseline, not an unmodified copy: interfaces were added and the density-of-states treatment is stated explicitly.

The re-parameterised baseline gives 18.99% (Voc 1.122 V, Jsc 19.68 mA/cm^2, FF 86.00%), within the published 17.9–19.06% band. Doping and defect floors remain modelling choices; measured doping is far lower [23]. Using the source paper's printed density of states instead gives 16.76%, so baseline replication depends materially on that choice.

TABLE II: Layer input parameters for the re-parameterised baseline. Screening subsequently moves ETL doping to 9×10^{17} cm^{-3} and explores ETL affinity; the ceiling values are in Table VI. Effective densities of states are $(N_c, N_v) = (2.2 \times 10^{18}, 1.8 \times 10^{19})$ cm^{-3} for CBTS, $(1.1 \times 10^{19}, 8.2 \times 10^{19})$ for CsPbI3, and $(2.0 \times 10^{18}, 1.8 \times 10^{19})$ for TiO2; contacts are Ni 5.5 eV (back) and ITO 4.0 eV (front) with $S_n = S_p = 10^7$ cm/s. All layers use equation-based Eg-sqrt absorption with a 10% front reflection filter. Parameters follow Hossain et al. [6] except for the added interfaces and the stated density of states.

Layer	d	E_g	χ	ε_r	Doping	N_t	μ_n/μ_p
CBTS HTL	100 nm	1.9	3.6	5.4	N_A 1e18	1e15	30/30
CsPbI3	800 nm	1.694	3.95	6.0	N_A 1e15	1e15	25/25
TiO2 ETL	30 nm	3.2	4.0	9.0	N_D 9e16	1e15	20/20

TABLE III: Interface defect layers at both heterojunctions: neutral single level, $\sigma = 10^{-19}$ m^2, $E_t = 0.6$ eV.

Interface	N (cm^{-2})	Sweep range
CBTS/CsPbI3	1e10	10^8–10^{12}
CsPbI3/TiO2	1e10	10^8–10^{12}

C Guardrails and simulation bases

Guardrails are part of the stated simulation box, not post-hoc filters. The bulk floor is 10^{12} cm^{-3} and the interface floor is 10^{10} cm^{-2}; points below the interface floor are audit points. Energetic screens reject Voc above the radiative bound or fill factor above Green's ideal-diode value.

Five bases recur in the paper and are defined once in Table V.

IV RESULTS AND DISCUSSION

A Grid screen and permissive-box ceiling

Round A screened thickness, acceptor density, bulk defects, ETL doping, and gap in one 96-run batch. Round B refined thickness, gap, and doping in 36 runs. The surveyed-affinity best reached 23.50%. Joint mapping of both transport-layer affinities then located the permissive-box ceiling at $\chi_{\mathrm{CBTS}} = 3.9$ and $\chi_{\mathrm{TiO2}} = 3.7$ eV.

Here 3.928 eV is the absorber affinity implied by the gap–affinity link at 1.65 eV, not an ETL affinity. The ETL affinity at the ceiling is 3.7 eV and the HTL affinity is 3.9 eV.

The ceiling sits at aggressive affinities outside surveyed values and at beyond-best-film defect density. It is therefore an upper anchor for the stated box, not a forecast. The JV curves are shown in Figure 2, and the thickness–defect interaction is shown in Figure 3.

B Band alignment

Figure 4 shows the ceiling alignment from input parameters. The absorber conduction edge lies below the TiO2 edge, while the absorber valence edge lies above the CBTS edge. Efficiency nevertheless peaks there.

Two readings can coexist. The interface activation-energy trend supports a recombination reading. The sharp efficiency fall at $\chi_{\mathrm{CBTS}} = 4.0$–$4.1$ eV is also consistent with a growing extraction barrier. The Ni work-function test constrains the contact regime, not the HTL/absorber valence offset, so it does not eliminate the barrier reading. Physical-plausibility flag: the ceiling sits at a 0.23 eV electron spike (absorber conduction edge -3.928 eV versus TiO2 -3.70 eV) and a 0.22 eV hole barrier (absorber valence -5.578 eV versus CBTS -5.80 eV) with a Ni contact, both outside surveyed affinity values and with contact cliffs that are partly SCAPS boundary behaviour. The 24.95%

TABLE IV: Guardrails and their anchors. "Assumption" means a modelling choice without a direct measurement behind it.

Guardrail	Value	Anchor
Bulk defect floor	$N_t = 10^{12}$ cm^{-3}	beyond-best-film lifetime guardrail
Interface floor	$N_{if} = 10^{10}$ cm^{-2}	assumption; demanding passivation regime
Capture cross-section	$\sigma = 10^{-19}$ m^2	assumption; swept $\times 0.1$–$\times 10$
Radiative coefficient	$C_r = 0$, priced at 10^{-10}/10^{-9} cm^3/s	assumption; literature bracket tested
Mobilities	25/25 absorber, priced	assumption; DFT-grade values tested
Parasitics	scripted R_s/R_{sh}	computed, not assumed zero
Contacts and affinities	stated and priced	surveyed versus adopted values separated
Optics	flat 10% reflection, edge-moving absorption	upper-envelope blue response; uncertainty carried
Screens	model-derived Voc and ideal-FF bounds	automated receipt gates; see Supplement

TABLE V: Simulation bases. Absorber affinity follows $\chi = 3.95 + 0.5(E_g - 1.694)$ unless a transport-layer affinity adoption is stated.

Base	Definition	Carries
Surveyed	Hossain parameters: $\chi_{\mathrm{CBTS}} = 3.6$, $\chi_{\mathrm{TiO2}} = 4.0$, $E_g = 1.694$, 0.8 μm	baseline 18.99%
Prior adopted	surveyed plus $N_D = 9 \times 10^{17}$, $E_g = 1.65$, $\chi_{\mathrm{TiO2}} = 3.8$	dose/audit anchor 23.84%
Ceiling	adopted plus $\chi_{\mathrm{CBTS}} = 3.9$, $\chi_{\mathrm{TiO2}} = 3.7$, 2.2 μm, $N_A = 3 \times 10^{16}$, $N_t = 10^{12}$	24.95% ceiling
Routine	surveyed transport affinities ($\chi_{\mathrm{CBTS}} = 3.6$, $\chi_{\mathrm{TiO2}} = 4.0$), $E_g = 1.65$, 1.0 μm, $N_A = 3 \times 10^{16}$, $N_D = 9 \times 10^{17}$, $N_t = 10^{15}$	reference 20.82%
Calibration	surveyed transport affinities, $E_g =$ 1.70, ITO 4.2 eV (dedicated def variant; stacked variants use ITO 4.4 eV), $R_s = 1$	ladder 19.33–20.66%

TABLE VI: Ceiling-point parameters and result. This is the highest evaluated guardrail-satisfying configuration on the named geometries, not a proven global optimum.

Layer	d	E_g (eV)	χ (eV)	Doping (cm^{-3})	N_t (cm^{-3})
CBTS HTL	100 nm	1.9	3.9	NA 1e18	1e15
CsPbI3 absorber	2.2 μm	1.65	3.928	NA 3e16	1e12
TiO2 ETL	30 nm	3.2	3.7	ND 9e17	1e15
Interfaces (both)	–	–	–	–	1e10 cm^{-2}
Result	Voc 1.321 V, Jsc 21.20 mA/cm^2, FF 89.11%, PCE 24.95%				

TABLE VII: Joint affinity map at guardrailed interfaces, PCE in percent. The peak turns over past $\chi_{\mathrm{CBTS}} \approx 3.95$ eV.

$\chi_{\mathrm{CBTS}} \setminus \chi_{\mathrm{TiO2}}$	3.7	3.8	3.9	4.0
3.50	22.10	22.10	22.10	22.07
3.60	23.83	23.82	23.80	23.50
3.70	24.67	24.64	24.53	23.77
3.80	24.89	24.86	24.67	23.81
3.90	**24.95**	24.91	24.74	23.81
3.95	24.94	24.90	–	–
4.00	24.81	24.79	–	–
4.10	23.25	23.25	–	–

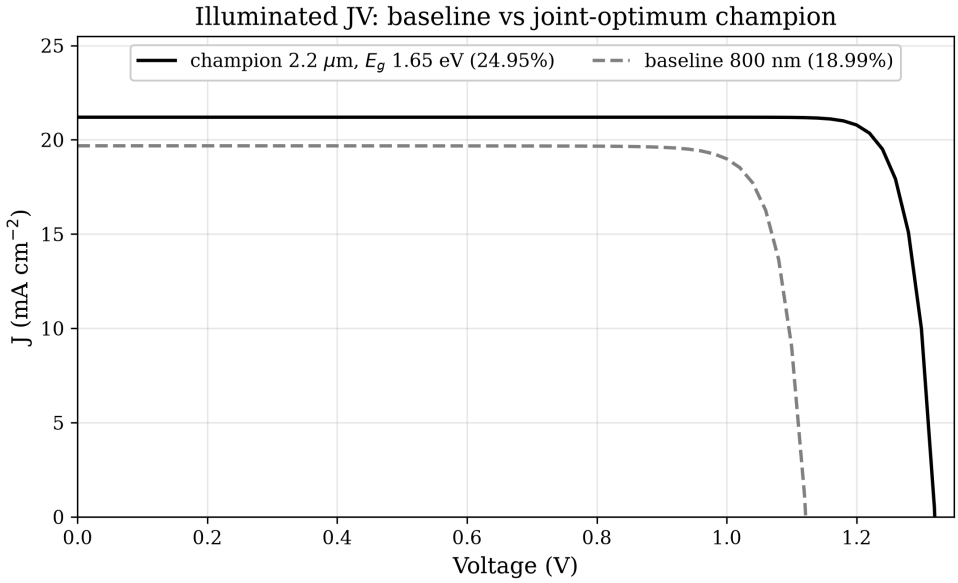


FIG. 2: Illuminated current–voltage curves: re-parameterised baseline versus permissive-box ceiling.

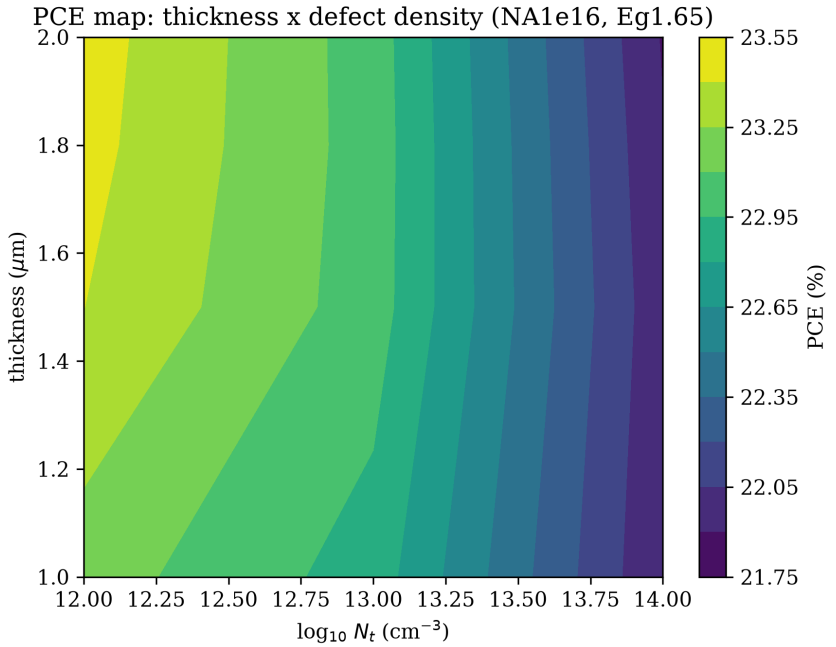


FIG. 3: Efficiency over absorber thickness and bulk defect density at surveyed transport affinities ($\chi_{\mathrm{CBTS}} = 3.6$, $\chi_{\mathrm{TiO2}} = 4.0$). The thickness optimum shifts with defect level.

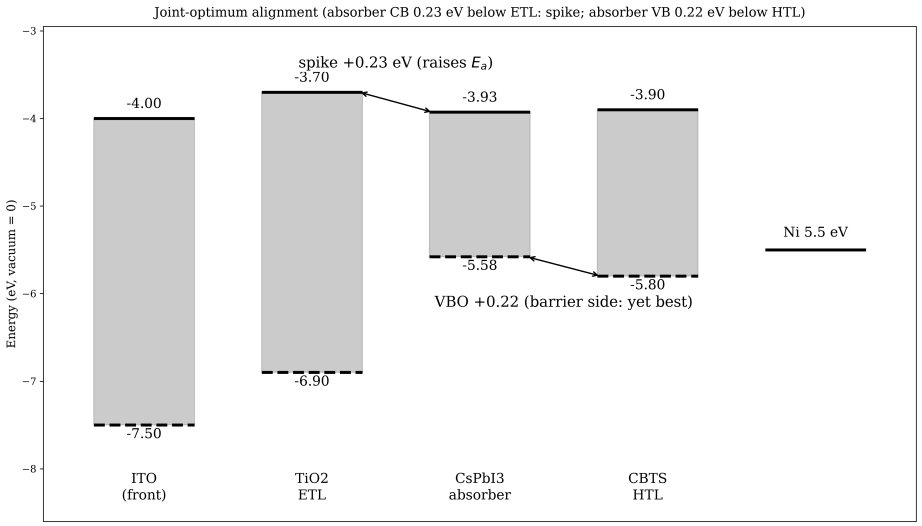


FIG. 4: Equilibrium band alignment of the ceiling stack from input parameters, energies versus vacuum.

value is therefore retained strictly as a permissive-box upper anchor; the defended number remains the 20.82% routine-defect result.

C Device characterisation

The QE edge moves from 731 to 750 nm between baseline and ceiling cells, supporting the Jsc gain in Figure 5. Because the flat blue plateau assumes no dispersive parasitic absorption, it is an upper envelope rather than a prediction.

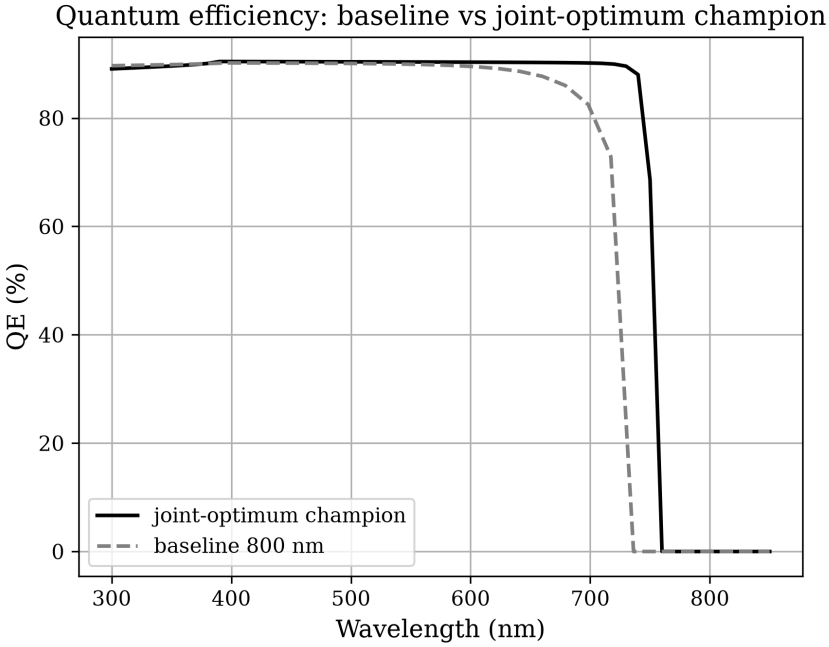


FIG. 5: Quantum efficiency of baseline versus ceiling cell; the edge shifts 731 to 750 nm.

The computed QE integral is 22.39 mA/cm^2 against the converged IV value of 21.20 mA/cm^2. For the stated optical stack, independently recomputed ideal step absorption gives 24.4 mA/cm^2 at 1.65 eV (23.4 at 1.70 eV, 19.8 at 1.80 eV; receipts `outputs/s18_analytic.json`); after the stated 10% front reflection the 1.65 eV ceiling is 21.96 mA/cm^2, below the QE integral. The IV value is therefore retained as the headlined current, with $\pm$1.2 mA/cm^2 carried as an absolute-scale systematic. Spectrum file substitution was blocked by the script interface, so independent spectrum discrimination was not achieved; a manual SCAPS GUI run with the spectrum file selected remains the proposed resolution (Supplement S4 records the blocked test).

Temperature dependence follows the established trend in Figure 6: efficiency falls from 25.88% at 275 K to 18.13% at 475 K, while Jsc remains near 21.20 mA/cm^2. The high-temperature end is a solver stress test because it omits gap narrowing and phase-stability effects.

D Priced radiative, mobility, and parasitic assumptions

Scripted radiative runs show that the zero-radiative default matters most at the clean corner. At the ceiling it costs 3.09 points at $C_r = 10^{-10}$ and 5.12 points at 10^{-9} cm^3/s. At routine defects the same values cost only 0.14 and 1.02 points. Mobility changes are negligible at the ceiling and 0.32 points at routine defects for DFT-grade 10/5 mobility [24].

Scripted parasitics give 24.73%, 24.52%, and 24.09% at $R_s = 0.5$, 1, and 2 Ωcm^2 on the ceiling base, extending to 23.66%, 23.23%, and 22.80% at $R_s = 3$, 4, and 5 Ωcm^2 (Table IX). At routine defects, $R_s = 1$ lowers 20.82% to 20.42%, reaching 19.62%, 19.22%, and 18.83% at $R_s = 3$, 4, and

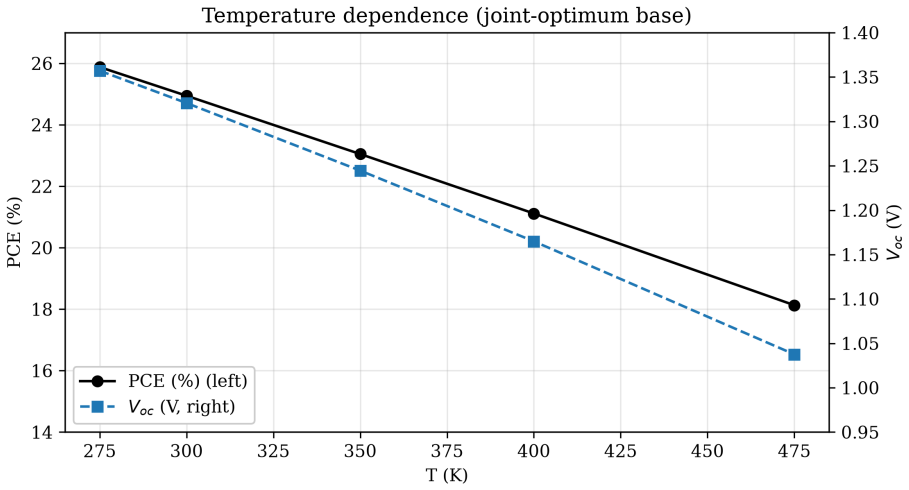


FIG. 6: Efficiency and open-circuit voltage versus temperature at the ceiling base. The 475 K end is a solver stress test.

TABLE VIII: Pricing the zero-radiative ($C_r = 0$) and mobility assumptions against the zero-radiative anchor of each base. C_r denotes the radiative recombination coefficient throughout.

Run	Voc (V)	Jsc (mA/cm^2)	PCE (%)	Δ (pts)
Joint optimum ($C_r = 0$)	1.321	21.20	24.95	–
$+C_r = 10^{-10}$ cm^3/s	1.163	21.08	21.86	−3.09
$+C_r = 10^{-9}$ cm^3/s	1.113	20.43	19.83	−5.12
μ_n/μ_p 10/5	1.321	21.20	24.94	−0.00
μ_n/μ_p 5/2	1.321	21.20	24.94	−0.01
Routine defects ($C_r = 0$)	1.147	20.77	20.82	–
$+C_r = 10^{-10}$ cm^3/s	1.141	20.73	20.68	−0.14
$+C_r = 10^{-9}$ cm^3/s	1.115	20.45	19.80	−1.02
μ_n/μ_p 10/5	1.148	20.60	20.50	−0.32

TABLE IX: Scripted series-resistance sweep on the ceiling base (2.2 μm, $N_t = 10^{12}$) and the routine base (1.0 μm, $N_t = 10^{15}$), surveyed affinities for the routine column. Receipts in `outputs/s36_rs.json` with $R_s = 0.5$–2 anchors in `outputs/s1_dose.json`.

R_s (Ωcm^2)	Ceiling PCE (%)	Ceiling FF (%)	Routine PCE (%)	Routine FF (%)
0	24.95	89.11	20.82	87.42
1	24.52	87.57	20.42	85.70
2	24.09	86.03	20.02	84.00
3	23.66	84.50	19.62	82.32
4	23.23	82.97	19.22	80.65
5	22.80	81.45	18.83	78.99

TABLE X: Record comparison under two density-of-states branches. Neither branch alone is presented as the unconditional physical result. The two physical-DOS values differ only in the hole density of states: $N_v = 4.1 \times 10^{18}$ cm^{-3} ($m_h^* = 0.3$) versus $N_v = 8.9 \times 10^{18}$ cm^{-3} ($m_h^* = 0.5$); N_c is unchanged. No-C_r rows fail the model-derived radiative screen and are excluded from ranking.

Branch	Ceiling treatment	PCE (%)	Voc (V)	Against 22.02%
Capped DOS, $C_r = 10^{-10}$	24.95 to 21.86	21.86	1.163	below record
Physical DOS ($m_h^* = 0.3$), $C_r = 10^{-10}$	J_dos03_cr10	23.46	1.241	above record
Physical DOS ($m_h^* = 0.5$), $C_r = 10^{-10}$	J_dos05_cr10	23.04	1.221	above record
Physical DOS ($m_h^* = 0.3$), no C_r	J_dos03	26.50	1.390	excluded
Physical DOS ($m_h^* = 0.5$), no C_r	J_dos05	26.11	1.373	excluded

5. Fill factor falls from 89.11% to 81.45% on the ceiling base and from 87.42% to 78.99% at routine defects across $R_s = 0$–5, so resistance alone closes roughly half of the six-point gap to measured fill factors near 83%; the remainder is attributed to nonradiative interface recombination (ideality above unity), for which illumination-dependent separation data are unavailable.

E Density-of-states conditionality

The density-of-states assumption controls the record comparison and is therefore stated prominently.

With capped density of states, literature radiative recombination lowers the ceiling to 21.86% (Voc 1.163 V), below the certified record. With effective-mass physical density of states, the same radiative term gives 23.46% at Voc 1.241 V ($N_v = 4.1 \times 10^{18}$ cm^{-3}) and 23.04% at Voc 1.221 V ($N_v = 8.9 \times 10^{18}$ cm^{-3}), above the record. Without radiative recombination, physical density of states gives 26.50% and 26.11%, but those runs fail the 1.371 V model-derived radiative screen (receipts `outputs/s26_diag.json`, `outputs/s31_small.json`, gates `outputs/s34_gates.json`). The capped branch preserves continuity across the campaign; the physical branch exposes the unphysical effective mass implied by the cap. The cap is therefore not defended as conservative physics when combined with radiative loss.

F Stacked realistic variants

Combining routine defects, literature radiative recombination, DFT-grade mobility, a 1.70 eV gap, ITO 4.4 eV, and $R_s = 1$ gives 19.33–19.59% across 0.6–1.0 μm (19.25% at $R_s = 2$ on 1.0 μm). This stacked range is below both the single-idealisation routine case and the certified record. It is an

TABLE XI: Stacked all-priced variants at surveyed affinities and routine defects ($E_g = 1.70$ eV, ITO 4.4 eV via the dedicated def variant, $C_r = 10^{-10}$ cm^3/s, mobility 10/5, R_s as listed).

Thickness	Voc (V)	Jsc (mA/cm^2)	FF (%)	PCE (%)
0.6 μm	1.195	18.93	85.45	19.33
0.8 μm	1.191	19.27	85.25	19.57
1.0 μm	1.188	19.38	85.09	19.59
1.0 μm, $R_s = 2$	1.188	19.38	83.60	19.25

TABLE XII: Calibration ladder at measured-grade inputs: $E_g = 1.70$, ITO 4.2 eV, $R_s = 1$, $C_r = 10^{-10}$, mobility 10/5, and $N_{if} = 3 \times 10^{10}$ unless noted.

Case	Voc (V)	Jsc (mA/cm^2)	FF (%)	PCE (%)
0.6 μm, $N_t = 10^{14}$	1.203	18.91	87.00	19.79
0.8 μm, $N_t = 10^{15}$	1.185	19.14	85.18	19.33
0.8 μm, $N_t = 3 \times 10^{14}$	1.198	19.26	86.47	19.94
0.8 μm, $N_t = 10^{14}$	1.203	19.29	86.90	20.16
0.8 μm, $N_t = 10^{13}$	1.205	19.31	87.13	20.27
1.0 μm, $N_t = 10^{14}$	1.202	19.47	86.83	20.32
0.8 μm, $N_t = 10^{14}$, $N_{if} = 10^{10}$	1.216	19.43	87.42	20.66

illustrative pessimistic corner, not an uncertainty interval.

G Measured-grade calibration

The model was driven toward measured-grade inputs without claiming a device-specific hindcast. The ladder climbs from 19.33% at $N_t = 10^{15}$ to 20.66% with interfaces at the 10^{10} floor, then saturates below the certified record.

The remaining gap to the certified record marks the distance between routine films and record films. Published device tables do not supply enough defect-level detail for a true device-specific hindcast, so none is claimed. As a forward check toward measured grades, vapour- and spin-coated film inputs (260 nm, N_t from reported lifetimes via $\tau = 1/\sigma v_{th} N_t$) give 17.96% and 16.76% respectively (receipts `outputs/s35_phys.json`): the correct order and plausible scale, but transfer error remains because interface velocities and doping for those specific devices are not reported. A fit that would carry weight requires, at minimum, Suns-Voc ideality data (to separate recombination from resistive loss) or injection-resolved TRPL lifetimes on absorber and half-stack films (to pin bulk τ and interface S independently); none of the record publications tabulated here supply both, so the ladder is reported as a transfer-error bound.

H Routine-quality sensitivity envelope

The prior volume was sampled with a 96-point Latin hypercube over eight factors (Table XIII; frozen in `priors.md`, seed 11). Held-fixed base: surveyed affinities, $E_g = 1.70$ eV with linked $\chi = 3.953$ eV, $N_A = 3 \times 10^{16}$ cm^{-3}, $N_{D,\mathrm{ETL}} = 9 \times 10^{17}$ cm^{-3}, $E_t = 0.6$ eV, $\sigma = 10^{-19}$ m^2, 300 K, 1 sun. The eight $C_r = 0$ runs form a separate sensitivity subset (seed 12) and are not pooled into the headline. Ninety-five in-prior runs converged.

The envelope median is 18.0%, with 5–95% range 15.6–20.0%. Bootstrap intervals are [17.7, 18.1] for the median, [15.3, 16.1] for p5, and [19.7, 20.2] for p95. The separate $C_r = 0$ median is 17.9%. Adding effective-mass density of states as another prior dimension moves the envelope only to 18.2%, with 15.8–20.0% range.

This is a prior-sensitivity envelope, not a confidence interval. Bulk and interface quality were sampled independently, although cleaner bulk and better interfaces are likely correlated in real films.

TABLE XIII: Frozen routine-quality prior ranges and their measured anchors. The $C_r = 0$ subset and the N_v density-of-states extension are reported separately, not pooled.

Factor	Prior	Anchor
$\log_{10} N_t$ (cm^{-3})	U(13, 16)	routine films $\sim 10^{15}$; intrinsic point defects $\sim 10^{14}$; 10^{12} floor priced separately
$\log_{10} N_{if}$ (cm^{-2}, both sides tied)	U(10, 13)	guardrail floor 10^{10}; flattens near 10^{13} at routine N_t
$\log_{10} C_r$ (cm^3/s)	U(-11, -9)	literature bracket; pessimistic top end (no photon recycling)
Mobility scale $\times 25/25$	U(0.2, 1)	DFT range 5–10 e^-/1–5 h cm^2/Vs (absorber)
R_s (Ωcm^2)	U(0.5, 3)	above the optimistic 0–2; ITO sheet resistance typically higher
ITO work function (eV)	{4.0, 4.2, 4.4}	archived def variants; 4.7 excluded (collapse regime)
Ni work function (eV)	{5.0, 5.2, 5.5}	archived def variants; joint-base cliff priced separately
Absorber d (μm)	U(0.4, 1.5)	experimental film window; 2.2 μm excluded (upper-bound geometry)
$\log_{10} N_v$ (cm^{-3}, extension)	U(18.6, 19.9)	hole mass 0.2–2.2; N_c fixed at 1.1×10^{19}

TABLE XIV: Guardrailed regime in portable metrics with $\sigma = 10^{-19}$ m^2 and $v_{th} = 10^5$ m/s: bulk lifetime τ and interface velocity S.

Density	τ / S	Note
$N_t = 10^{12}$ cm^{-3} (floor)	$\tau = 100$ μs	beyond-best-film regime
$N_t = 10^{13}$ cm^{-3}	$\tau = 10$ μs	–
$N_t = 10^{14}$ cm^{-3}	$\tau = 1$ μs	–
$N_t = 10^{15}$ cm^{-3} (routine)	$\tau = 0.1$ μs	routine case
$N_{if} = 10^{10}$ cm^{-2} (floor)	$S = 100$ cm/s	floor of stated box
$N_{if} = 10^{11}$ cm^{-2}	$S = 1000$ cm/s	–

The 2024–2026 records fall above the envelope; earlier lower-efficiency results do not. The gap therefore measures what record quality requires beyond these priors.

I Interface dose–response

Removing the interface blocks inflates the surveyed-base cells by 1.26–1.66 points, almost entirely through Voc. The full dose–response runs from 25.11% at 10^8 cm^{-2} to 21.20% at 10^{12} cm^{-2} on the prior-adopted base. At routine defects the response is nearly flat: 10^9–10^{11} cm^{-2} moves efficiency only from 20.84% to 20.67%. Interfaces therefore dominate at the defect floor, while bulk defects dominate at routine defect density.

No-interface Voc values sit at or very near the model-derived radiative bound because the model omits a radiative channel. Those cells are therefore disqualified by lack of margin, not ranked.

TABLE XV: Identical cells with interfaces ON versus OFF.

Cell	IF ON (%)	IF OFF (%)	Inflation (pts)
1.5 μm cell	23.40	24.87	+1.47
2.0 μm cell	23.47	24.73	+1.26
Surveyed-base best	23.50	25.16	+1.66

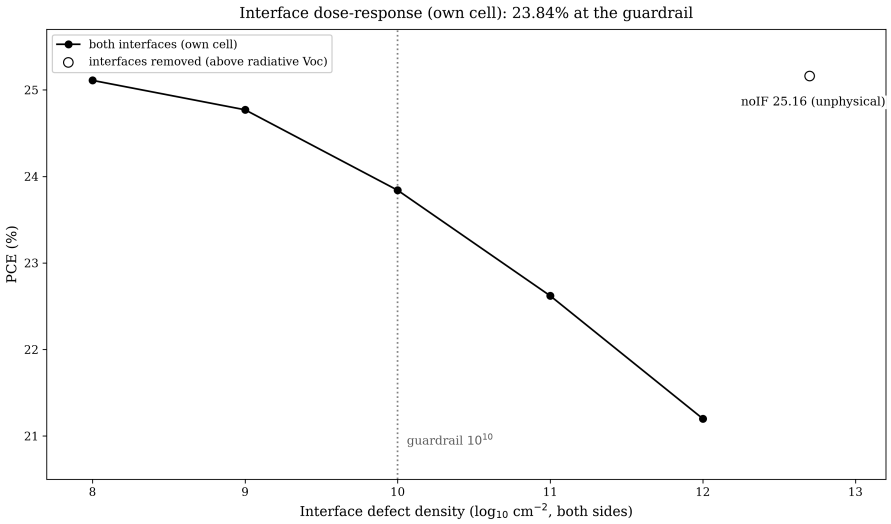


FIG. 7: Efficiency versus interface density on the prior-adopted base ($\chi_{\mathrm{TiO2}} = 3.8$, $E_g = 1.65$). The no-interface point (25.16%) is the surveyed-base audit value from Table XV, shown for scale as an audit point, not a candidate.

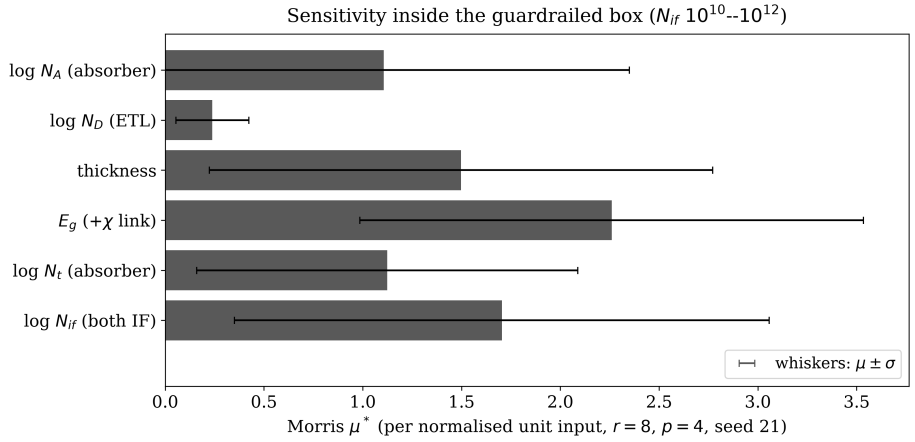


FIG. 8: Morris μ^* with interaction spread σ. Rankings are regime conditional and use different physical units across factors.

J Morris sensitivity

A guardrail-clean Morris screen used eight trajectories, four levels, six factors, and 56 converged runs spanning 15.34–22.96%. Per decade, interfaces lead at 0.85 points against 0.37 for bulk defects, but their bootstrap intervals overlap. The sharper paired-trajectory difference is +0.58 points with interval [-0.59, 1.97], with five of eight trajectories positive. Morris alone therefore does not establish the interface lead.

The regime conclusion instead rests on the dose–response and the nominal S versus d/τ anchors [25]. Those anchors organise the clean-corner and routine-defect regimes; binning the prior-volume runs by the same ratio does not reveal a clean internal crossing at this sample size. The same nominal interface velocity can also recombine differently as alignment changes the interface carrier densities.

TABLE XVI: Loss budget: permissive-box ceiling against measured devices and model limits. The Lei row mixes printed and headline values as stated in its source.

Device	Voc (V)	Jsc (mA/cm^2)	FF (%)	PCE (%)
This work (box ceiling)	1.321	21.20	89.11	24.95
This work + $C_r = 10^{-10}$	1.163	21.08	89.13	21.86
This work, stacked realistic range	1.19	18.9–19.4	83.6–85.5	19.25–19.59
AP-modified CsPbI3 [18]	1.29	20.77	83.01	22.16
QD-interface CsPbI3 [19]	1.28	–	–	22.15
Dual-interface CsPbI3 [4]	1.27	–	–	21.71
Carbon HTL-free [26]	1.202	20.72	83.2	20.72
Model-derived radiative bound, this stack (1.65 eV)	1.371	24.4 (step-absorption ref.)	90.5	–

K Loss budget and limits

The ceiling exceeds the certified record by 2.93 points at 1.65 eV (24.95 versus 22.02%) and by 2.71 points even at 1.70 eV (24.73%, Voc 1.364 V, Jsc 20.23 mA/cm^2, FF 89.58%; receipts `outputs/s36_rs.json`). The excess cannot be assigned to the gap alone. Simulated fill factor near 89% exceeds measured values near 83%, alongside permissive contacts and affinities.

The approximately six-point fill-factor discrepancy (89.11% simulated versus ∼83% measured) is the largest remaining gap and is now bounded rather than unresolved. Scripted $R_s = 3$–5 Ωcm^2 lowers ceiling fill factor to 84.50–81.45% and routine fill factor to 82.32–78.99% (Table IX), so measured series resistance accounts for roughly half the gap at plausible R_s values. A missing back reflector affects Jsc rather than fill factor, so it is not offered as the explanation. The remainder is attributed to nonradiative interface recombination (ideality above unity); ideality-factor and illumination-dependent data sufficient to separate the two losses exactly are unavailable, so the decomposition is reported as a bound.

This is one stack, one solver, and one dimension. Several contact cliffs are SCAPS boundary behaviour as much as device physics. The checklist below (Table XVII) is therefore a proposed pilot reporting standard, not an established publication standard.

V CONCLUSION

This study uses SCAPS-1D simulation to evaluate the performance of CsPbI3 solar cells on the ITO/TiO2/CsPbI3/CBTS/Ni stack under interface-inclusive guardrails. Joint optimisation of thickness, doping, defects, gap, and both transport-layer affinities locates a permissive-box ceiling of 24.95% (Voc 1.321 V, Jsc 21.20 mA/cm^2, FF 89.11%; 2.2 μm absorber, N_A 3×10^{16} cm^{-3}, N_t 10^{12} cm^{-3}, χ_{CBTS} 3.9 / χ_{TiO2} 3.7 eV). Sampling a frozen routine-quality prior volume gives a prior-predictive envelope of median 18.0% with 5–95% range 15.6–20.0%, above which the 2024–2026 experimental records fall, and the defended routine-defect result is 20.82% (20.42% at $R_s = 1$ Ωcm^2). Every load-bearing assumption is priced: defect floors, gap, contacts, affinities, radiative recombination, mobility, parasitics to $R_s = 5$ Ωcm^2, optics, and density of states, for which the ceiling-versus-record comparison is reported both ways. The principal remaining discrepancy is fill factor, bounded as approximately half resistive and half recombination-driven. Although the simulated ceiling reflects the theoretical potential of the stack, the defect floor sits beyond the best reported films and both optimum affinities lie outside surveyed values, so reproducing such performance experimentally requires strict oxidation control, effective defect passivation, and optimised interfaces.

TABLE XVII: Pilot guardrailed-reporting checklist for comparable SCAPS claims. The full protocol mapping is in Supplement S7.

Item	Disclose	This work	Protocol
Interfaces	status + N_{if} + σ or S	10^{10} cm^{-2}, 10^{-19} m^2	reporting
Bulk defects	N_t + σ or τ	10^{12} cm^{-3}; routine 10^{15}	reporting
Radiative coefficient	C_r value used	0, priced at 10^{-10}/10^{-9}	inputs
Mobilities	μ_n/μ_p plus anchor	25/25, priced at 10/5, 5/2	inputs
Units and scripts	script versus file systems	Supplement S2	reporting
Parasitics	R_s/R_{sh} basis	scripted; routine case included	inputs
Contacts and affinities	work functions and surveyed/adopted χ	priced	inputs
Optics and current	reflection, absorption, spectrum, QE/IV treatment	uncertainty carried	spectra
Radiative screen	model-derived Voc bound and deviations	1.371 V at 1.65 eV (in-model; independent step-absorption Jsc 24.4 in `outputs/s18_analytic.json`)	radiative check
Receipts	definitions plus raw outputs	from the corresponding author upon reasonable request, Supplement S1	reporting

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Conflict of interest

The authors have no conflicts to disclose.

Author contributions

Hussain Touhid Siddiquee: conceptualization, methodology, software, formal analysis, investigation, data curation, visualization, validation, and writing—original draft, review and editing. Md. Abdul Malek Fahim: software, validation, and writing—original draft.

Data availability

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Note on AI use: AI assistance was used during revision for editing, consistency checking, and reference-label correction. All SCAPS results come from the archived simulation receipts and were not generated by AI.

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